

Total No. of Questions :6]

SEAT No. :

P65

OCT. -16/BE/Insem. - 119

[Total No. of Pages :3

B.E. (Mechanical Engineering)

RELIABILITY ENGINEERING

(2012 Pattern) (Semester - I) (Elective - I)

Time : 1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) *All questions are compulsory i.e. Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data, if necessary.*
- 4) *Use of electronic pocket calculator is allowed.*
- 5) *Neat diagrams must be drawn wherever necessary.*

Q1) a) Calculate failure density and hazard rate for the failure data of 400 Allen screws tested simultaneously for 240 hrs. **[6]**

Time Interval (hrs.)	0-40	40-80	80-120	120 -160	160-200	200-240
Number of failed components	114	70	30	30	60	96

b) Define and distinguish between reliability and quality of the product. **[4]**

OR

Q2) a) Twelve gaskets of an I.C. engine were tested for life and following results were recorded for analysis. Find the mean time to failure for 12 gaskets. Also find mean failure rate for $T = 268$ hrs and 300 hrs. **[6]**

specimen Number	1	2	3	4	5	6	7	8	9	10	11	12
Time to Failure hrs	205	214	225	231	243	249	254	261	268	276	285	300

b) Explain the term warranty management. Comment on the statement "Improper warranty management may cost manufacturer in terms of loss of profits". **[4]**

P.T.O.

- Q3) a)** Five elements (a, b, c, d and f) of a system are connected as shown in Fig. 1 which also indicates the reliability of each component. Find the system reliability using cut - set method and also draw an equivalent block diagram using minimal cut - set elements. [6]

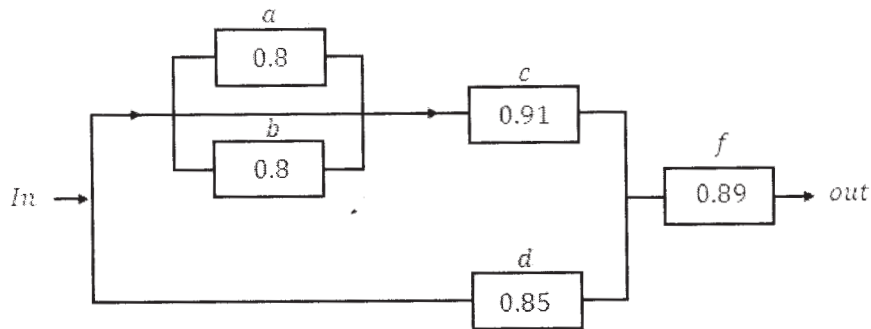


Figure 1

- b) Write in brief about [4]
- Mutually exclusive events,
 - Conditional probability

OR

- Q4) a)** The reliability value of components 1, 2, 3, 4, 5 and 6 as shown in Fig. 2, are $R_1 = 0.89$, $R_2 = 0.9$, $R_3 = 0.95$, $R_4 = 0.85$, $R_5 = 0.95$ and $R_6 = 0.96$ respectively. Find the reliability of the system assuming all the components are independent. Also, find reliability of the system when components are identical having reliability of $R = 0.99$ and independent. [6]

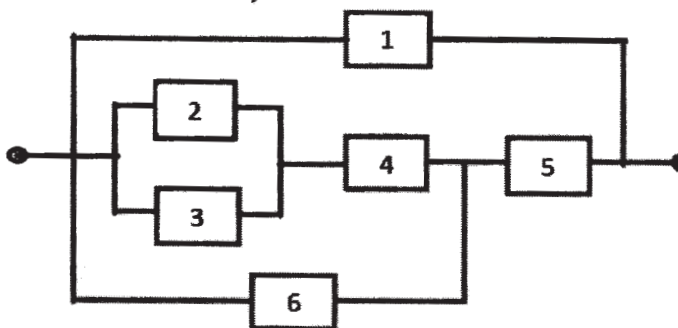


Figure 2

- b) Compare the Normal and Log-normal probability distributions. [4]

- Q5)** a) A system of three elements 1, 2 and 3 are having failure rates $\lambda_1 = 0.007$, $\lambda_2 = 0.003$, $\lambda_3 = 0.001$ per hour respectively. Find failure rates as well as reliability of each sub system for the entire mission period using ARINC apportionment technique assuming mission time of 30 hours and desired system reliability of 0.91. [6]
- b) Write about what is meant by redundancy, its types and its use in system with the help of sketches. [4]

OR

- Q6)** a) An electronic unit consists of three resistors having reliabilities of 0.65, 0.72, 0.86 connected in series. The reliability of an electronic unit is desired as 0.60. Find for which resistors the reliability values are to be improved and also find the values of individual reliabilities of the resistors by using minimum effort method. [6]
- b) Explain equal apportionment technique and state its limitations. [4]

